

MARKING SCHEME

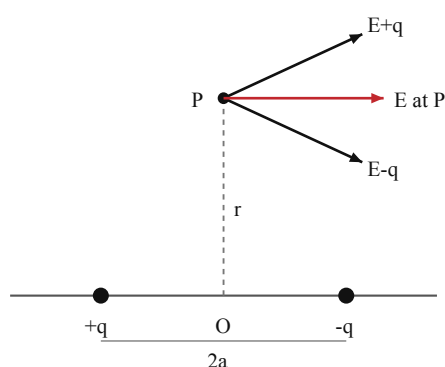
PHYSICS (042) - Class XII, 2026 Q.P. Code 55/1/1

Section E, questions 31 to 33 (5 marks each). Value points and distribution of marks.

General instructions to examiners: award marks step-wise as indicated. Where a candidate uses any other correct method, award full marks. In questions carrying an internal choice, mark the alternative actually attempted.

Q31 (main option) - electric dipole

Deriving an expression for the electric field E due to a dipole at a point on the equatorial plane	2 1/2
Writing the expression for electric field at a far off point	1/2
Calculating the force and torque	2



(a) Labelled figure showing the dipole and the equatorial point P 1/2

The magnitudes of the electric fields due to the two charges +q and -q are given by

$$E_{+q} = (1/4 \pi \epsilon_0) \cdot q / (r^2 + a^2) \quad 1/2$$

$$E_{-q} = (1/4 \pi \epsilon_0) \cdot q / (r^2 + a^2)$$

The components normal to the dipole axis cancel away. The components along the dipole axis add up. The total electric field is opposite to p. 1

$$E = -(E_{+q} + E_{-q}) \cos(\theta) \hat{p}$$

$$E = -(1/4 \pi \epsilon_0) \cdot 2qa / (r^2 + a^2)^{3/2} \cdot \hat{p} \quad 1/2$$

At large distance $r \gg a$

$$E = -2qa / (4 \pi \epsilon_0 r^3) \cdot \hat{p} \quad 1/2$$

(b) Net force on the dipole

$$F = F_{+q} + F_{-q} = [+q(2 \hat{i}) - q(2 \hat{i})] \quad 1/2$$

$$F = 0 \text{ N} \quad 1/2$$

$$\text{Torque } \tau = \vec{p} \times \vec{E} = p(-\hat{i}) \times 2 \hat{i} \quad 1/2$$

$$\tau = 0 \quad 1/2$$

Alternatively: $\tau = pE \sin(\theta)$. The angle between $\vec{p}$ and $\vec{E}$ is π , hence $\tau = 0$. Award the same marks.

Total: 5

Q31 (OR option) - cells in parallel

Deriving the expression for equivalent e.m.f.	2
Deriving the expression for equivalent internal resistance	1
Finding the current through resistance $2R$	2

(a) With $I = I_1 + I_2$ and V the common terminal potential difference

$$V = E_1 - I_1 r_1 \text{ and } V = E_2 - I_2 r_2 \quad 1/2 + 1/2$$

$$I = (E_1 - V)/r_1 + (E_2 - V)/r_2$$

$$V = (E_1 r_2 + E_2 r_1)/(r_1 + r_2) - I (r_1 r_2 / (r_1 + r_2)) \quad 1/2$$

$$\text{Comparing with } V = E_{eq} - I r_{eq} : E_{eq} = (E_1 r_2 + E_2 r_1)/(r_1 + r_2) \quad 1/2$$

$$r_{eq} = r_1 r_2 / (r_1 + r_2) \quad 1$$

(b) With $E_1 = E$, $E_2 = 3E$ and $r_1 = r_2 = R$

$$E_{eq} = (ER + 3ER)/(R + R) = 4ER/2R = 2E \quad 1/2 + 1/2$$

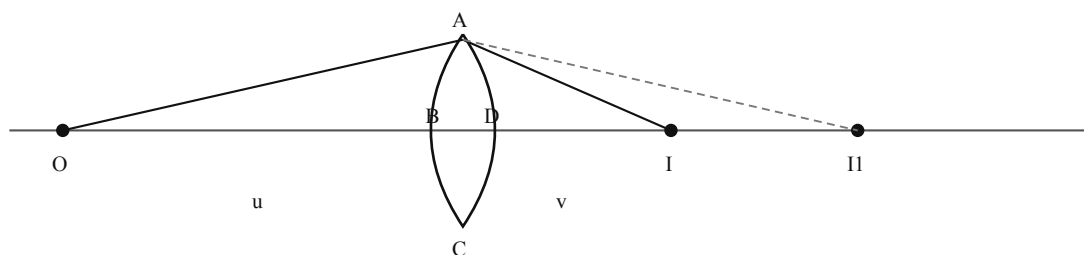
$$r_{eq} = R/2 \quad 1/2$$

$$I = E_{eq} / (2R + R/2) = 4E / 5R \text{ A} \quad 1/2$$

Total: 5

Q32 (main option) - lens maker's formula

Deriving the expression for lens maker's formula	3
Finding the distance of the final image from the object	2



(a) Labelled ray diagram for refraction at the two surfaces of a thin lens

1

The first refracting surface forms the image of the object O at I_1 . For refraction from the first interface ABC

$$n_1/OB + n_2/B I_1 = (n_2 - n_1)/BC1 \quad \dots(1) \quad 1/2$$

Similarly for refraction from the second interface ADC, the image I_1 acts as a virtual object for the second surface

$$-n_2/D I_1 + n_1/DI = (n_2 - n_1)/DC2 \quad \dots(2) \quad 1/2$$

For a thin lens $B I_1 = D I_1$. Adding equations (1) and (2)

$$n_1/OB + n_1/DI = (n_2 - n_1) [1/BC1 + 1/DC2]$$

$$-n_1/u + n_1/v = (n_2 - n_1) [1/R_1 - 1/R_2]$$

$$1/v - 1/u = (n_2/n_1 - 1) [1/R_1 - 1/R_2] \quad 1/2$$

If the object is kept at infinity the image will form at the focus, hence

$$1/f = (n_2/n_1 - 1) [1/R_1 - 1/R_2] \quad 1/2$$

(b) For lens L_1 , with $u = -80$ cm and $f = 40$ cm

$$1/40 = 1/v_1 + 1/80 \text{ gives } v_1 = 80 \text{ cm} \quad 1/2$$

For lens L2: $u_2 = 120 - 80 = 40$ cm, so $1/v_2 = 0$ and $v_2 = \text{infinity}$	1/2
For lens L3: $u_3 = \text{infinity}$, so $v_3 = 40$ cm	1/2
Distance between final image and object = $80 + 120 + 20 + 40 = 260$ cm	1/2

Total: 5

Q32 (OR option) - concave mirror and mirror formula

Drawing the ray diagram	1
Deriving the mirror formula	2
Calculating the focal length of the mirror	2

(a) Ray diagram for an object between F and C of a concave mirror 1

Triangles BAP and B'A'P are similar: $BA/B'A' = AP/A'P$... (1) 1/2

Triangles MNF and B'A'F are similar, and since N is very close to P with $BA = MN$: 1/2

$BA/B'A' = PF/A'F$... (2)

From (1) and (2): $-u/-v = -f/(-v + f)$ 1/2

$1/f = 1/v + 1/u$ 1/2

Award full marks if the student derives the mirror formula by any other justified method.

(b) For a two times magnified virtual image, $m = 2$

$m = -v/u$, so $v = -2u = -2(-10) = 20$ cm 1/2 + 1/2

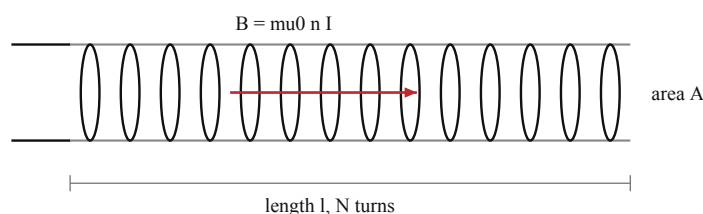
$1/f = 1/20 - 1/10 = (1 - 2)/20$ 1/2

$f = -20$ cm 1/2

Total: 5

Q33 (main option) - Faraday's law, self-inductance, rotating rod

Stating Faraday's law	1
Deriving the expression for self-inductance	2
Finding the induced e.m.f.	2



(a) The magnitude of the induced e.m.f. in a circuit is equal to the time rate of change of magnetic flux through the circuit. Alternatively, $e = -d(\phi_B)/dt$. 1

(b) Magnetic field due to a current carrying long solenoid of length l and area of cross section A having n turns per unit length is $B = \mu_0 n I$

Total flux linked: $N \phi_B = (n l)(\mu_0 n I)(A) = \mu_0 n^2 A l I$ 1/2 + 1/2

$L = N \phi_B / I$ 1/2

$L = \mu_0 n^2 A l$ (equivalently $L = \mu_0 N^2 A / l$) 1/2

Award full marks for any other correct alternative method.

(c) Induced e.m.f. of a rod pivoted at one end

$e = (1/2) B l^2 \omega$	1/2
$e = (1/2) \times 4 \times 10^{-3} \times (50 \times 10^{-2})^2 \times (2\pi \times 1)$	1
$e = 3.14 \text{ mV}$	1/2

Note: 60 rpm = 1 revolution per second, so $\omega = 2\pi \text{ rad/s}$. No mark for the substitution step if this conversion is not made.

Total: 5

Q33 (OR option) - step-up transformer

Drawing a labelled diagram of a step-up transformer	1
Stating the principle	1/2
Obtaining the ratio of voltages in terms of number of turns and current	2
Finding (i) current in the primary coil	1/2
Finding (ii) output voltage	1

(a) Labelled diagram showing primary, secondary and soft iron core 1

Principle: when an alternating voltage is applied to the primary, the resulting current produces an alternating magnetic flux which links the secondary and induces an e.m.f. in it (mutual induction). 1/2

$e_s = -N_s d(\phi)/dt \quad \dots(1)$	1/2
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$e_p = -N_p d(\phi)/dt \quad \dots(2)$	1/2
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From (1) and (2): $v_s/v_p = N_s/N_p$	1/2
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For an ideal transformer $I_p V_p = I_s V_s$, hence $V_p/V_s = I_s/I_p$	1/2
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(b) With $N_p : N_s = 1 : 5$, $P = 5 \text{ kW}$ and $V_p = 200 \text{ V}$

(i) $I_p = P/V_p = 5000/200 = 25 \text{ A}$	1/2
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(ii) $V_s = 5 \times 200 = 1000 \text{ V}$	1
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Total: 5

Transcribed from the official CBSE marking scheme for Physics (042), Q.P. code 55/1/1, 2026. Greek letters and vector notation from the original are spelled out here so the text layer stays searchable.